

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6 HT	(a)			Indicative content: While food is stored or transported the growth of bacteria must be slowed down or stopped by: • refrigeration – slows down but does not stop bacterial growth • freezing – stops bacteria multiplying but does not kill them • heating – heating then rapid cooling kills nearly all microbes • cooking – kills microorganisms • drying – removes water so bacteria cannot digest and absorb the food source • salting – bacteria lose water from cells so they dehydrate and cannot reproduce • pickling – addition of vinegar to lower pH and inactivate microorganisms. 5– 6 marks Very good even coverage of precautions and storage options linked with effect on bacteria – at least 5 methods linked <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3 – 4 marks Good coverage with some correct linking (at least two methods explained) <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i>	6			6		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				1-2 marks Simple list of precautions/one precaution linked <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate used limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks No attempt made or no response worthy of credit.						
	(b)	(i)		1:10 000 (1) CFU have not clumped together but there are (>20 therefore) enough (1)			2	2		
		(ii)		total number of CFU ignoring anomaly = 171 (1) mean number = $\frac{171 (ecf)}{5} = 34.2$ (1) Count = $34.2 (ecf) \times 100\,000 = 3\,420\,000$ per 0.1 cm^3 (1) So number of bacteria = $34\,200\,000(ecf) / \text{cm}^3$ (1) If anomaly not ignored then (3) max for answer of 32 000 000		4		4	4	
				Question 6 total 12	6	4	2	12	4	0